

Valdis Slokenbergs

Curriculum Vitae

Texas Tech University — Lubbock, TX — vslokenb@ttu.edu

EDUCATION

Ph. D. in Physics: Texas Tech University — 2024 - present

M.S. in Physics: Texas Tech University — 2025

B.S. in Physics: Johns Hopkins University — 2020 - 2024

RESEARCH ACTIVITIES

TTU High Energy Physics group

2024 - present

Investigating electroweak vector boson scattering in the semileptonic channel at the *CMS* experiment using data collected during Run 2 and Run 3 of the *LHC*.

- Developing signal/background discriminator using XGBoost and advanced machine learning algorithms (ex. transformers) for signal sensitivity optimization
- Developing and maintaining analysis workflows within PocketCoffea analysis framework
- Using data-driven method to model the radiative QCD contribution to background processes.
- Deriving custom shape corrections to various jet taggers for input to machine learning algorithms.

Participating in *CMS Particle Flow* development for the *High Luminosity Large Hadron Collider (HL-LHC)*.

- Studied performance of existing reconstruction methods on special high pileup dataset, to assess ability to extrapolate to future detector environment [1].

Participating in instrumentation studies based at TTU Advanced Particle Detector Lab.

- Electronics and software commissioning, development of online TTU database, and shipping contact for silicon module assembly for the Phase 2 *CMS High Granularity Calorimeter* upgrade.
- Participated in the test beam campaign of the high granularity dual-readout calorimeter prototype designed at TTU.

Preparing for dark sector upgrade to the fixed-target experiment *SpinQuest* at *Fermilab*. *Delayed due to FNAL PIP-II upgrade*.

- Reanalyzing commissioning dataset to extract preliminary dark sector sensitivity.
- Led effort to demonstrate particle identification using shower shape with new electromagnetic calorimeter and test beam data [2].
- Main student on critical studies of track reconstruction and trigger road upgrade which demonstrate no significant performance degradation post dark sector upgrade.

JHU Experimental Particle Physics group

2021 - 2024

Undergraduate researcher studying offshell Higgs boson production at *CMS* and anomalous couplings at *FCC-ee*.

- Performed generator validation for offshell Higgs boson production using JHUGen.
- Constrained effect of CP-violation to HVV anomalous couplings using kinematic observables at *FCC-ee* [3].

CERN Summer Student

June 2023 - August 2023

Study of recoil uncertainty in top mass measurement at *CMS*.

- Implemented novel recoil feature from *Pythia* generator into shared *CMS* software.
- Evaluated contribution to top quark mass uncertainties [4].

SELECTED PUBLICATIONS

- [1] CMS Collaboration. *Performance of jet, missing transverse momentum, and (machine-learned) particle flow reconstruction in pp collision data with more than 100 pileup interactions at the CMS experiment*. CMS-DP-2026-111. July 2026. URL: <https://cds.cern.ch/record/2967023>.
- [2] Aram Apyan et al. “Performance measurements of the electromagnetic calorimeter and readout electronics system for the DarkQuest experiment”. In: *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1080 (Nov. 2025), p. 170792. ISSN: 0168-9002. DOI: 10.1016/j.nima.2025.170792. URL: <http://dx.doi.org/10.1016/j.nima.2025.170792>.
- [3] Nicholas Pinto et al. *Constraining CP-violating contributions in the Higgs-strahlung process at FCC-ee*. Aug. 2025. DOI: 10.17181/839bh-ckm18. URL: <https://doi.org/10.17181/839bh-ckm18>.
- [4] V. Slokenbergs. *Recoil uncertainty in top mass measurements*. Aug. 2023. URL: <https://repository.cern/records/byvhs-yqr93>.

PRESENTATIONS

Constraining CP-odd contributions in the Higgs-strahlung process at FCC-ee using kinematic observables

APS April Meeting 2024: Particle Physics I, April 2024.

Recoil uncertainty in top mass measurement

CERN Summer Student Sessions, August 2023.

SCHOLARSHIPS

AT&T Chancellor’s Fellowship: Texas Tech University — Years: 2024-2028

Bucy Graduate Scholarship in Applied Physics: Texas Tech University — 2025, 2026

Arthur and Catherine B. Adel Scholarship: Johns Hopkins University — 2023, 2024

TEACHING

Texas Tech University

Graduate teaching assistant

Intermediate Physics Lab - Spring 2025, Spring 2026
Modern Physics Lab - Fall 2025
Principles of Physics I Lab - Fall 2024

Johns Hopkins University

Undergraduate Learning Assistant

General Physics I & II - Summer 2021, Summer 2022
General Physics I Lab - Fall 2021

PUBLIC ENGAGEMENT

Media

A glimpse of tomorrow’s detector, today

Highlighted for CMS public briefing. August 2026.

Texas Tech Physics Students Test Detector Prototype at CERN

Interviewed for *Texas Tech Now* article. November 2025.

Texas Tech Doctoral Student Looks to Have an Impact in Particle Physics

Interviewed for *Texas Tech Now* article. May 2025.

Latvieši sevi piesaka CERN, jeb jauno zinātnieku mobilitāte [Latvians at CERN, or young scientist mobility]

Interviewed for radio broadcast *Globālais latvietis. 21. gadsimts [The global Latvian in the 21st century]*, Latvijas Radio 1. October 2023.

Community Service

69th South Plains Regional Science and Engineering Fair

Judge reviewing local elementary and middle school students scientific research activities and posters. February 2025.